AIRBNB PRICE PREDICTION USING LINEAR REGRESSION

Minor Project 2 | New York City Airbnb Open Data (2019)

1. Objective

The objective of this project is to explore Airbnb listing data and build a machine-learning model that estimates listing prices using listing characteristics such as location, room type, minimum nights, reviews and availability.

2. Dataset and Preprocessing

The dataset contains 48,895 New York City Airbnb listings and 16 original columns. Duplicate records were removed, unnecessary identifier/text fields were dropped, and missing values in **reviews_per_month** were filled with 0. The remaining dataset contained 11 columns and no missing values. IQR-based capping was applied to **price** and **minimum_nights** to limit extreme values while retaining the rows.

3. Feature Engineering and Model

The target variable was **price**; the remaining columns were used as input features. Categorical variables (**neighbourhood_group**, **neighbourhood** and **room_type**) were converted using One-Hot Encoding. The data was divided into 80% training and 20% testing sets using random_state=42. A Linear Regression model was trained through a preprocessing pipeline. Residual plots and the distribution of residuals were also examined.

4. Model Evaluation

Metric	Result	Interpretation
R^2 Score	0.5235	About 52.35% of target-price variation explained
Mean Absolute Error (MAE)	$41.80	Average absolute prediction error
Mean Squared Error (MSE)	3,277.86	Average squared prediction error
Root Mean Squared Error (RMSE)	$57.25	Error scale with greater penalty for large errors

5. Cheaper vs. Expensive Listings

The average absolute error was **$30.09** for cheaper listings and **$53.52** for expensive listings. Therefore, the model had lower dollar prediction error for cheaper listings and found expensive listings more difficult to predict.

6. Conclusion

The Linear Regression model achieved an R^2 score of 0.5235, indicating moderate predictive performance on the processed target. The results show that listing prices can be estimated from the available features, although substantial prediction error remains. More advanced models, additional location-related features and careful treatment of skewed prices may improve performance.

Methodological note: IQR capping was performed before the train-test split. Consequently, the reported metrics measure performance on the capped price target rather than the original uncapped listing prices. For a stricter evaluation in a future revision, calculate capping limits on the training data only and apply those limits to the test data.

Tools: Python | Pandas | NumPy | Matplotlib | Seaborn | Scikit-learn | Google Colab